

Aishwarya Kumar Arvind

+1(425)518-0080 | Email | LinkedIn | Portfolio | GitHub

Software Developer with a Master of Science in Computer Science from Northeastern University. Designs, develops, and maintains secure web applications, building interactive web interfaces backed by well-structured server-side code and application databases, with a focus on secure data handling, performance, scalability, accessibility, and cross-browser delivery.

EXPERIENCE

National Benefit Services (NBS)

Oct 2025 - Present

Software Developer | Next.js, React, Tailwind, TypeScript, Python, Flask, REST APIs, OAuth/OIDC, Azure, SQL Denver, CO / Remote

- Design, develop, and maintain a secure enterprise contract management web application for a third-party benefits administrator, delivering contract intake, review workflows, document tracking, and administrative reporting through interactive web interfaces.
- Build responsive, accessible web interfaces using Next.js, React, Tailwind, and TypeScript; evaluate code to ensure proper structure, standards compliance, usability, and compatibility across browsers and devices.
- Implement web authentication and access-control flows using OAuth/OIDC (MSAL, Azure AD/B2C) for secure handling of sensitive participant data, incorporating privacy and accessibility considerations.
- Develop real-time web collaboration features using WebSocket and Yjs, and write supporting server-side code and RESTful APIs in Python and Flask for contract workflow automation, business logic, and data validation.
- Maintain the SQL databases and Microsoft Azure components that support the web system, and build front-end web tests with Playwright and Vitest to improve performance, scalability, reliability, and release quality.

Healthcare Systems Engineering Institute (Northeastern University)

Jul 2023 - Oct 2025

Research Assistant - Full Stack Web Developer | Python, Flask, JavaScript, React, AWS, DynamoDB

Boston, MA

- Designed and developed a full-stack healthcare web application using a Flask backend and React/JavaScript frontend to support hospital operations, epidemic surge planning, and real-time decision-support workflows.
- Developed and maintained the interactive web interfaces and RESTful APIs that support the web application, integrating DynamoDB for efficient data retrieval and scalable web workflows.
- Configured AWS infrastructure (Nginx, Gunicorn, CloudFront) and implemented GitHub Actions CI/CD pipelines to improve web application performance, reliability, and iterative releases within an Agile Scrum team.
- Developed web-based visualization and presentation features to communicate hospital resource-allocation outcomes to healthcare professionals during epidemic surge scenarios.

Accenture

May 2018 - Feb 2021

Software Engineer | ASP.NET/C#, MySQL, PySpark, Unix Shell Scripting

Mumbai, India

- Developed enterprise software and backend solutions and collaborated with full-stack development teams to integrate backend pipelines with user-facing applications, improving usability and decision-support capabilities.
- Designed and deployed automation that improved team productivity by 150+ hours per month and contributed to client revenue recovery initiatives; recognized with the APEX Excellence Award.

RESEARCH

Cybersecurity at MIT Sloan (CAMS)

Jun 2024 - Oct 2025

Research Volunteer | Python, NetworkX, Gurobi, Data Visualization

Boston, MA

- Developed software tools and data-visualization workflows to model organizational cyber resilience, critical-infrastructure dependencies, and restoration strategies after disruption events, using Python, NetworkX, and Gurobi.

Co-author, AHFE International Conference (Poster Presentation) | Honolulu, HI

Dec 2025

- Development and Usability of Tools to Improve Hospital Resiliency to Capacity Surges* - presented healthcare systems engineering research (Healthcare Systems Engineering Institute, NEU); developer of the web-based visualization and decision-support tool.

PROJECTS

AI-Assisted Resource Navigator - Web Application | TypeScript, React, Python, OpenAI API

Sep 2024 - Dec 2024

- Developed an HCI-focused web application using TypeScript, React, and Python to support guided queries, verified resource discovery, and document summarization, achieving 85% task completion in usability testing.

QUANS - Web Application | JavaScript, HTML, CSS, React, Redux, Node.js, MongoDB

Sep 2022 - Dec 2022

- Designed and developed a collaborative discussion-forum and knowledge-market web application using React, Redux, Node.js, and MongoDB, integrating multiple web APIs to improve platform performance, data flow, and user-driven Q&A functionality.

TECHNICAL SKILLS

Web Development: ReactJS, Next.js, Tailwind, Redux, Node.js, Flask, HTML/CSS, REST/JSON APIs, OAuth/OIDC (MSAL, Azure AD/B2C), WebSocket/Yjs, Responsive Design, Cross-Browser Testing, Accessibility (WCAG 2.2 AA), Secure Web Application Development

Languages: JavaScript, TypeScript, Python, SQL, NoSQL, HTML/CSS

Databases: SQL Server, PostgreSQL, MySQL, MongoDB, DynamoDB

Cloud and DevOps: Microsoft Azure, AWS (EC2, S3, CloudFront, DynamoDB), Docker, GitHub Actions, CI/CD

Testing and Practices: Playwright, Vitest, Agile/Scrum, Jira, Git, Data Validation, Secure Web Application Development

EDUCATION

Northeastern University, Houry College of Computer Sciences

Boston, MA

Master of Science in Computer Science; GPA: 3.95/4.00

Sep 2022 - Dec 2024

PSG College of Technology

Coimbatore, India

Bachelor of Engineering in Electronics and Communication

Jun 2014 - May 2018

ACHIEVEMENTS

APEX Excellence Award (Individual), Accenture PLC | *recognized for client impact and contract profitability*

2019